

This note describes the problem with a lender of last resort (LOLR). We assume one-period debt. Notation is very similar to the previous note.

## 1 Model

Value of no default.

$$\begin{aligned}
v^{nd}(b, \ell, g, s, \epsilon) &= \max_{c, n, \ell'} \left\{ \frac{c^{1-\gamma}}{1-\gamma} + \beta g^{1-\gamma} \mathbb{E} \left[ \max \left\{ v^{nd}(b', \ell', g', s', \epsilon'), v^d(b', \ell', g', s', \epsilon') \right\} | g, s \right] \right\} \quad (1) \\
c + b + \ell &= y + n + n^\ell \\
y &= g e^{\sigma \epsilon} \\
g b' &= R(n, \ell', g, s) n, \quad g \ell' = R_{nd}^\ell n^\ell \\
\ell' &\leq \bar{\ell}_{nd}, \quad n \leq \bar{n}(g)
\end{aligned}$$

where  $\ell$  is the amount of debt outstanding with the LOLR, and  $n^\ell$  is the new issuance with the LOLR. We assume the LOLR charges a given (gross) rate  $R_{nd}^\ell$ . There is a limit  $\bar{\ell}_{nd}$  on how much the sovereign can borrow from the LOLR. The LOLR rate and borrowing limit depend on default status but not on other states—we'll make a function of  $s$  later on.

We are assuming that the schedule from private lenders depends on  $\ell$ : that is,  $R(n, \ell', g, s)$ . This implies a timing assumption: the sovereign borrows from the LOLR first, and then borrows from private markets. [LETS DISCUSS ABOUT THIS]

Value of default

$$\begin{aligned}
v^d(b, \ell, g, s, \epsilon) &= \max_{c, n^\ell} \left\{ \frac{c^{1-\gamma}}{1-\gamma} + \beta g^{1-\gamma} \mathbb{E} \left[ \theta v^d(b', \ell', g', s', \epsilon') \right. \right. \\
&\quad \left. \left. + (1 - \theta) \max \left\{ v^{nd}(\kappa b', \ell', g', s', \epsilon'), v^d(b', \ell', g', s', \epsilon') \right\} | g \right] \right\} \quad (2) \\
c + \ell &= \phi(g) y + n^\ell \\
y &= g e^{\sigma \epsilon} \\
g b' &= b, \quad g \ell' = R_d^\ell n^\ell \\
\ell' &\leq \bar{\ell}_d
\end{aligned}$$

The assumption is that the sovereign cannot default on the LOLR. We further assume that the sovereign can also keep on borrowing from the LOLR during periods of default/market exclusion. However, the LOLR can set a rate  $R_d^\ell$  and a borrowing limit  $\bar{\ell}_d$  that are different during periods of default/market exclusion.

Let  $\mathbf{b}'(b, \ell, g, s, \epsilon)$  be the optimal borrowing policy,  $\ell'_{nd}(b, \ell, g, s, \epsilon)$  be the borrowing policy from LOLR during no default, and  $\ell'_d(b, \ell, g, s, \epsilon)$  be the borrowing policy from LOLR during default.

Schedule

$$1 = \frac{R(n, \ell', g, s)}{1 + r^*} \mathbb{E}[Q(b', \ell', g', s' \epsilon') | g, s] \quad (3)$$

$$gb' = R(n, \ell', g, s)n \quad (4)$$

with  $\ell' = \ell'_{nd}(b, \ell, g, s, \epsilon)$  in equation (4).

Let  $z = \{g, s, \epsilon\}$  collect all the exogenous sates. Prices are given as

$$Q(b, \ell, z) = [1 - \mathbf{d}(b, \ell, z)] + \mathbf{d}(b, \ell, g, \epsilon)X(b, \ell, z) \quad (5)$$

$$X(b, \ell, z) = \frac{1}{1 + r^*} \mathbb{E} \left[ \theta X(b', \ell'_d(b, \ell, z), z') + (1 - \theta) \times \right. \\ \left. \{ [1 - \mathbf{e}(b', \ell'_d(b, \ell, z), z')] X(b', \ell'_d(b, \ell, z), z') + \mathbf{e}(b', \ell'_d(b, \ell, z), z') \kappa Q(\kappa b', \ell'_d(b, \ell, z), z') \} | g \right] \quad (6)$$

where  $\mathbf{d}(\cdot)$  and  $\mathbf{e}(\cdot)$  are the default and re-entry policies, and  $b' = b/g$  in equation (6). IMPORTANT: to solve equation (6) we need the borrowing policies:  $\mathbf{b}'(b, \ell, z)$ ,  $\ell'_{nd}(b, \ell, z)$ ,  $\ell'_d(b, \ell, z)$ . The borrowing policy  $\mathbf{b}'(b, \ell, z)$  is implied by the optimal issuance policy and the schedule as in equation (1).

The default and re-entry decisions,  $\mathbf{d}(b, \ell, g, s, \epsilon)$  and  $\mathbf{e}(b, \ell, g, s, \epsilon)$ , are given as

$$\mathbf{d}(b, \ell, g, s, \epsilon) = \begin{cases} 0, & \text{if } v^{nd}(b, \ell, g, s, \epsilon) \geq v^d(b, \ell, g, s, \epsilon), \\ 1, & \text{otherwise;} \end{cases} \quad (7)$$

$$\mathbf{e}(b, \ell, g, s, \epsilon) = \begin{cases} 1, & \text{if } v^{nd}(b, \ell, g, s, \epsilon) \geq v^d(b, \ell, g, s, \epsilon), \\ 0, & \text{otherwise.} \end{cases} \quad (8)$$

## 1.1 Implementation Comments

We need to set the borrowing limits  $\{\bar{\ell}_{nd}, \bar{\ell}_d\}$  and the rates  $\{R_{nd}^\ell, R_d^\ell\}$ . Let's start assuming the borrowing limit and the rates do not depend on the market status:  $\bar{\ell}_{nd} = \bar{\ell}_d = \bar{\ell}$  and  $R_{nd}^\ell = R_{nd}^\ell = R^\ell$ .

For the borrowing limit. Let  $\bar{g} = \mathbb{E}[g]$  be the average growth rate (on the ergodic distribution). I would set  $\bar{\ell} = \Delta \bar{g}$ . Start with small numbers, moving from 1% to 5%. That is  $\Delta \in \{0.01, 0.02, 0.03, 0.04, 0.05\}$ .

For the interest rate. Take the risk-free rate and add a few points. Say from 4 p.p. to 10 p.p. So we would have  $R^\ell = 1 + r^* + \sigma^\ell$  and  $\sigma^\ell = \{\frac{4}{100}, \frac{6}{100}, \frac{8}{100}, \frac{10}{100}\}$ . We can explore more values for  $\sigma^\ell$  later on. Ideally, we want to make sure  $\sigma^\ell$  is large enough so that it's not subsidizing borrowing.

## 1.2 Solution algorithm

Step 0: Set grids. Grid for debt,  $\vec{b} = \{b_1, b_2, \dots, b_{N_b}\}$ , with  $b_1 < 0$  (saving) and  $b_{N_b} > 0$  (borrowing). Grid for LOLR lending  $\vec{\ell} = \{\ell_1, \ell_2, \dots, \ell_{N_\ell}\}$ . Grid for endowment shocks,  $\vec{\epsilon} = \{\epsilon_1, \epsilon_2, \dots, \epsilon_{N_\epsilon}\}$ .

I would start with small values for  $N_b$  and  $N_\ell$ . Probably around 1000 for each. Once it's running, we will want to increase the grid sizes.

For  $\vec{\epsilon}$  let's do the same as in the paper:  $N_\epsilon = 17$  and use Tauchen method. (We should move to Rouwenhorst method. See here: <https://www.karenkopecy.net/RouwenhorstPaper.pdf>).

We also need  $\vec{g} = \{g_L, g_H\}$ , the transition matrix  $P_g(g'|g)$ , and the variance  $\sigma_\epsilon$ . All these numbers come from the estimation (see Table 1 in our paper).

Step 1: Guess  $v^{nd}(b, \ell, g, s, \epsilon)$ . Also guess borrowing policies from the LOLR during no default:  $\ell'_{nd}(b, \ell, g, s, \epsilon)$ .

Step 2: Given  $v^{nd}(b, \ell, g, s, \epsilon)$ , solve for  $v^d(b, \ell, g, s, \epsilon)$ . Note that, given  $v^{nd}(b, \ell, g, s, \epsilon)$ , equation (2) is a fixed point in  $v^d(b, \ell, g, s, \epsilon)$ . Iterate on equation (2) until  $v^d(\cdot)$  converges.

From this step, save  $\ell'_d(b, \ell, g, s, \epsilon)$

Step 3: Given  $v^{nd}(b, \ell, g, s, \epsilon)$  and  $v^d(b, \ell, g, s, \epsilon)$ , compute  $\mathbf{d}(b, \ell, g, s, \epsilon)$  and  $\mathbf{e}(b, \ell, g, s, \epsilon)$  as in equations (7)-(8).

Step 4: Given the borrowing policy  $\ell'_d(b, \ell, g, s, \epsilon)$ , the default and re-entry decisions,  $\mathbf{d}(b, \ell, g, s, \epsilon)$  and  $\mathbf{e}(b, \ell, g, s, \epsilon)$ , and our guess for borrowing policies  $\ell'_{nd}(b, \ell, g, s, \epsilon)$ , we solve for the prices  $Q(b, \ell, g, s, \epsilon)$  and  $X(b, \ell, g, s, \epsilon)$  using equations (5)-(6).

In particular, guess  $X(b, g, s, \epsilon)$ . Then compute the price  $Q(b, g, s, \epsilon)$  using equation (5), and the implied  $X(b, g, \epsilon)$  using equation (6). Iterate until  $X(\cdot)$  converge.

Step 5: We need to compute the schedule  $R(n, \ell', g, s)$ . This is the hardest step, and even more now with two debt choices. The computations below is for a given  $g$ .

Consider debt levels for next period:  $b' = b_i$  and  $\ell' = \ell_j$ . That is, private debt is the  $i^{\text{th}}$  position of the grid  $\vec{b}$  and LOLR debt is the  $j^{\text{th}}$  position of the grid  $\vec{\ell}$ .

Compute the expected price tomorrow associated with these decisions  $b' = b_i$  and  $\ell' = \ell_j$ . That is:  $Q_{ij}^{\mathbb{E}}(g, s) = \mathbb{E}[Q(b_i, \ell_j, g', s', \epsilon')|g, s]$ . The implied rate then is  $R_{ij}(g, s) = \frac{1+r^*}{Q_{ij}^{\mathbb{E}}(g, s)}$ . The issuance then is  $n_{ij}(g, s) = \frac{gb_i}{R_{ij}(g, s)}$

We can do this  $\forall i = 1, \dots, N_b$  and  $\forall j = 1, \dots, N_\ell$ . The implied schedule is a collection of points as  $\tilde{\mathcal{S}}(g, s) = \{R_{ij}(g, s), n_{ij}(g, s)\}_{i,j}$ .

Compute expected default probability  $p_{ij}(g, s) = \mathbb{E}[\mathbf{d}(b_i, \ell_j, g', s', \epsilon')|g, s]$ . We will impose an upper bound  $p_{ub}$  on schedule choices based on this default probability. In particular, take the points of the schedule with default probability below  $p_{ub}$ :  $\mathcal{S}(g, s) = \left\{ \{R_{ij}(g, s), n_{ij}(g, s)\}_{i,j} : p_{ij}(g, s) \leq p_{ub} \right\}$ .

Step 6: We need to determine which parts of the schedule are available to the borrower depending on the sunspot realization,  $s$ .

In the good sunspot,  $s = s_g$ , we allow the borrower to choose from all of the schedule  $\mathcal{S}(g, s_g)$ . (Keep in mind that we already eliminated the high-default probability choices).

In the bad sunspot,  $s = s_b$ , we select points with two criteria: (1) has to be an increasing part of the schedule, and (2) for a given issuance, if there is a higher rate in an increasing part of the

schedule, we jump to the higher rate.

Step 7: Now we update  $v^{nd}(b, \ell, g, s, \epsilon)$  and  $\ell'_{nd}(b, \ell, g, s, \epsilon)$ . For given current states  $\{b, \ell, g, s, \epsilon\}$ , we do as follows.

Given our current guess  $v^{nd}(\cdot)$ , compute the value  $w(b, \ell, g, s, \epsilon) = \max \{v^{nd}(b, \ell, g, s, \epsilon), v^d(b, \ell, g, s, \epsilon)\}$ , and the expected value  $w^{\mathbb{E}}(b, \ell, s, g) = \mathbb{E}[w(b, \ell, g', s', \epsilon')|g, s]$ .

That is, for each  $b' = b_i \in \vec{b}$  and  $\ell' = \ell_j \in \vec{\ell}$ , evaluate the value of the policy

$$\begin{aligned}\hat{v}_{ij}^{nd}(b, \ell, g, s, \epsilon) &= \frac{c_{ij}^{1-\gamma}}{1-\gamma} + \beta g^{1-\gamma} w^{\mathbb{E}}(b_i, \ell_j, g, s) \\ c_{ij} &= g e^{\sigma \epsilon} + n_{ij}(g, s) + n_j^\ell - b - \ell \\ n_j^\ell &= \frac{g \ell_j}{R^\ell}\end{aligned}\tag{9}$$

Then, we update our guess

$$\hat{v}^{nd}(b, \ell, g, s, \epsilon) = \max_{i,j} \left\{ \hat{v}_{ij}^{nd}(b, \ell, g, s, \epsilon) \text{ s. to } \ell_j \leq \bar{\ell} \text{ and } n_{ij} \in \mathcal{S}(g, s) \right\}\tag{10}$$

When computing the max in equation (10), we need to impose that the issuance choice is within the schedule that corresponds to the sunspot  $s$ . In practice, when  $s = s_b$ , we impute a very negative pay-off for choices that are not in the schedule.

This optimization will also give a new borrowing policy from the LOLR:  $\hat{\ell}'(b, \ell, g, s, \epsilon) = \ell_j$ .

Compare  $\hat{v}^{nd}(b, \ell, g, s, \epsilon)$  and  $\hat{\ell}'(b, \ell, g, s, \epsilon)$  to the initial guess  $v^{nd}(b, \ell, g, s, \epsilon)$  and  $\ell'(b, \ell, g, s, \epsilon)$ . If it's close, we are done. Otherwise, update  $v^{nd}(b, \ell, g, s, \epsilon)$  and  $\ell'(b, \ell, g, s, \epsilon)$ , and go back to *Step 2*.